

Forage: Understanding RAG-based Sensemaking for Community Conversations

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Abstract

We introduce FORAGE, a RAG-based and LLM-augmented search engine, which we apply to the problem of sensemaking for community conversation data. We report on formative user studies introducing FORAGE to two distinct user groups: NPR journalists and municipal staff in the city of Durham, North Carolina. We taxonomize the query types users make with the tool, use cases that include synthesizing insights across conversations and finding content about a particular subject. We find that users tend to gravitate towards using the system for synthesis more than for pure search. We report on challenges and opportunities surfaced by performing sensemaking with an open-ended interface like FORAGE, such as the benefits of finding content quickly, but also the challenges users face interacting with a system in natural language. Insights from this formative study confirm the usefulness of FORAGE for sensemaking, but also make follow-up work, such as systematically evaluating system performance and developing appropriate design, urgent.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

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1 Introduction

Retrieval-augmented generation (RAG) has become a popular component of information retrieval systems, and they power large-scale web search engines like Google's AI Overviews and small-scale personal search systems such as NotebookLM. They are prevalent in mainstream commercial use and show potential for assisting in querying small-scale data, but few RAG-based tools are publicly accessible, and little is currently known about user needs or user behavior when interacting with them. "Sensemaking"—or "the action or process of making sense of or giving meaning to something" [2]—may be supported by RAG capabilities. For example, RAG-based systems could help users make flexible queries in English to retrieve content, generate summaries, and even inform thematic analysis of complex content. Without a foundational understanding of how individuals who perform sensemaking work may want to use these tools, we lack the information to robustly evaluate the performance of LLMs serving in these functions, or the information necessary to design tools to facilitate appropriate use. In this work, we begin to address this gap through a formative study that observes how participants interact with our RAG-based sensemaking tool, a prototype called FORAGE¹. We presented 19 study participants with an opportunity to interact with FORAGE as a tool in a data analysis workflow within two distinct user groups: journalists, and municipal workers.

Our research questions were:

- How do users naturally interact with open-ended RAG-based systems such as FORAGE when performing sensemaking work?
- When interacting with FORAGE, how do users validate content?

¹Repository: <https://github.com/mit-ccc/forage-public>

- How might RAG-based sensemaking tools balance the advantages of open-endedness with an interface that supports appropriate system use?

In this work, we offer three main contributions. First, we contribute the introduction of the FORAGE prototype. Second, we conduct and present results on a formative study to understand the needs and interests of two user groups looking to explore and make sense of community-centric data, and we report on their experiences with and interest in using the FORAGE prototype. Third, following several months of user interaction with FORAGE, we review user logs and taxonomize common patterns of querying behavior on FORAGE. This, in combination with our participant interviews, offers insight into the common uses of open-ended RAG systems for users interacting with community-centric data, including decreasing effort required to sort through long hours of conversations or assistance in generating thematic insight and synthesis. These insights surface opportunities and challenges for the design of RAG-based sensemaking systems, which are on the verge of ubiquity. Our findings can help inform design and future evaluation of similar systems.

2 Related Works

The era of large language models (LLMs) has opened up new possibilities for interacting with information systems through conversational user interfaces (CUIs) like ChatGPT. As LLMs suddenly rose in public discourse following ChatGPT's release, their drawbacks, including their tendency towards "hallucination," have been noted with concern. One technical approach to mitigate hallucination is through retrieval-augmented generation (RAG) systems [17], which can enhance "accuracy and credibility of the generation, particularly for knowledge-intensive tasks" [12]. RAG-based systems offer a promising prospect: they combine the open-ended, naturalistic interaction users have come to appreciate about LLM CUIs, with the potential to ground query responses on a particular source of data that can be verified. Large-scale consumer-facing products, like the recently released OpenScholar [6], use this idea to allow users to query a large database in natural language. For example, one could query "Can you suggest 3 papers that study LLMs' abilities of generating new research ideas in NLP?" and the system produces a synthesized response that links to citations in the Semantic Scholar database. Google also now provides users with "AI Overviews" of searched topics, which are frequently presented above traditional search results. Traditional search paradigms are giving way to new ones that provide contextually-relevant synthesis atop retrieved data using LLMs.

The term "sensemaking" has many definitions, but it generally refers to the process of making sense of and assigning meaning to complex, ambiguous information [30]. Jeong and Brower identify three stages in sensemaking: noticing, interpretation, and action [16, 31]. When data is collected for the purposes of social research, or for understanding a group of individuals, their insights, and concerns, the process of analyzing the data includes "sensemaking." This can include both formal and informal approaches to exploring the data, identifying patterns, naming them, systematically annotating them, and understanding their meaning in the context of their origins.

LLMs are increasingly used as sensemaking and analysis tools across tasks in research and practice. Journalists view AI entering the newsroom with both excitement and trepidation, seeing its potential for efficient data processing, but also noting concerns about accuracy, bias, and worry for the future of the profession [10, 19, 22]. Governments are also grappling with these changes. Older natural language processing (NLP)-based methods, including topic modeling, have been part of some workflows within city governments that try to make sense of open-ended public comments [14] and other forms of public participation with government [24]. Many text processing tasks are useful in the governmental context, including summarizing public comments, discovering and grouping citizen contributions by topic, understanding public sentiment, and identifying stakeholders [24]. However, some methods, such as statistical topic modeling, require technical skill to implement, which not all parties in government may have [29]. Traditional keyword searches often fail to surface higher-level thematic connections or contextual nuances, whereas LLMs interest users with their flexibility in generating thematic insight and synthesis. LLMs also open up the possibility of flexible analysis to new kinds of users because less technical knowledge is now needed to interact with LLM-based interfaces compared to past methods in NLP.

To that point, recent work shows that qualitative researchers are interested in using LLM-based tools to explore data, brainstorm, transform data, and assist in analysis [26]. Ongoing work explores opportunities and risks LLMs bring to text analysis, including the process of making sense of open-ended surveys [25], which could support analysis of stakeholder input in the governmental and organizational context. On the other hand, a growing body of work raises concerns about the potential for LLMs to homogenize knowledge production by introducing biased defaults and making some kinds of knowledge production, but not others, tractable [18].

Commercial RAG-based systems that operate on web and scholarly data have quickly entered common use, without much insight thus far as to their application towards private, small-scale data, or their ability to support sensemaking without code. As such, little is publicly known about how these open-ended but data-grounded systems might be used in sensemaking, what users might want or expect from them, and how their appropriateness for a variety of tasks may be further evaluated in order to inform responsible development. This gap in understanding motivates our formative study observing user interactions with a RAG-based sensemaking system in two community sensemaking contexts. We share the first known data characterizing user logs of RAG queries on community-specific data, and we contextualize these findings with in-depth user interviews centering on our prototype, FORAGE.

3 FORAGE System

FORAGE is a retrieval-augmented generation search engine designed to help researchers explore a large database of conversation data. In its prototype configuration, it indexes transcribed speaker turns from the Fora project [27], a corpus of facilitated small-group dialogues hosted within diverse communities on topics of community concern such as education, public health, and housing affordability. FORAGE employs a four-stage pipeline. First, a *retrieval* step identifies the speaker turns that are most relevant to the user's natural

language query by comparing text embeddings of the query (embedded using all-MiniLM-L6-v2 from HuggingFace) against a pre-computed embedding index of all speaker turns in the corpus, which are stored in a vector database for efficient nearest-neighbor search. Next, a selection step takes the top 100 speaker turns returned by retrieval. These candidate passages are then input, along with the original user query, into a large language model (LLM) prompt, which produces a synthesized "analysis" according to instructions given by the query, such as *"Find the stories of immigrants coming to the US."* On the backend of FORAGE, a prompt then instructs the LLM to cite specific speaker turn IDs in its generated analysis, which allows the final text to reference the most relevant excerpts.² Finally, the system re-sorts the retrieved results, ranking the results in descending order based on how frequently each retrieved conversation turn is cited in the LLM-generated analysis. This allows the most thematically salient speaker turns to appear more prominently in displayed search results, shown on the left-hand side of the results page, with the LLM-generated analysis appearing for users on the righthand side of the page. An example screenshot of FORAGE's query response is available as Figure 4 in the appendix.

In our first deployment of FORAGE, diagrammed in Figure 1, users interacted with the system through an interface consisting of a single field of input in free text, posing concrete or abstract queries such as *"jobs involving animals"* or *"themes related to gentrification."* In our second deployment of FORAGE, users interacted with the system by decomposing their query into an **Objective + Subject** scheme, such as **Objective:** *Generate themes about* + **Subject:** *voter disenfranchisement*. In this version of the system, we use the provided **Subject** only as the query embedding for retrieval, and both the **Objective + Subject** to prompt the LLM. This update is shown in Figure 3, with motivations and implications discussed later.

4 Methods and Data Collection

In order to understand user interests in RAG-based systems applied to community data, we deployed a two-part user study that engaged two distinct sets of participants, each reflecting a sense-making context in the public interest space. We collaborated with Cortico, a US-based non-profit that designs and provides technology to support civic dialogue, and has active collaborations with National Public Radio ("NPR"), an American public broadcasting organization and national syndicator to a network of local public radio stations across the US, and with the city of Durham, North Carolina ("City of Durham") to host small group conversations in the style of facilitated dialogues in the Fora Corpus [27]. These collaborations each produced a series of recorded conversations, which NPR and the City of Durham could then use to understand insights surfaced by participants. The data from these conversations form the data input to FORAGE for participants to query later in our interview study.

4.1 Community conversation data

Collaborating with Cortico, NPR recruited a core set of 10 journalists working at distinct member stations across the US to host

conversations about the topic of "community." In the NPR conversation collection, 10 journalists at local radio stations across the United States hosted conversations with community members in that location, totaling 28 conversations over the course of 2 months in the fall of 2024, conversations which produced 38 hours of audio data. Groups of three to six individuals from the community were invited to share their stories in places like Orlando, Florida, western Colorado, and Kansas City, Kansas. Each community participant in the NPR conversation collection was asked to respond to the same set of open-ended questions about their sense of community, what they wanted to learn about others, and ways they felt their community was misunderstood [23]. Participants in these recorded community conversations shared responses with the knowledge that their answers would be recorded and used for journalistic and research purposes. The NPR journalists conducting conversations then produced radio stories about community across cities, a task which required analyzing long hours of the recorded conversation data.

Similarly, an ongoing community engagement project at the City of Durham hosts community conversations in partnership with Cortico in order to solicit personal stories from citizens, particularly those from underrepresented and disproportionately impacted communities, and inform future directions for the city. Members of the Community Engagement Division of Durham managed a large series of 70 facilitated dialogues that surfaced community stories and feedback spanning multiple divisions of the city government beginning in 2022. Participants in these conversations shared their voices with the intention that their perspectives and stories would be recorded and heard by the city, and available through a public-facing platform called BeHeard Durham. These conversations hosted by the City of Durham were ongoing through 2024 and into 2025, and 70 had occurred by the time of this study. After recording, conversations were then subject to a manual sensemaking process that involved team members performing qualitative annotation by reviewing each conversation. The City of Durham partners expressed interest in tools that could easily help search, sort, and synthesize the growing corpus of conversation data so that it could be more easily used to inform other aspects of the civic process.

4.2 Participant needs

In the case of both the NPR team and the City of Durham, individuals working to analyze the conversations, or "sensemakers," were reviewing and annotating the conversation data manually, but were frustrated with the slow speed and tedium of manual annotation. Given the size of the conversation corpora each had under review: 38 hours across 28 NPR conversations and 64 hours across 70 Durham conversations, manual review was a mounting challenge. Given this analysis challenge, the partners had been seeking additional tools to support conversation sensemaking beyond the analysis platform offered through Cortico. These groups learned about the FORAGE prototype through the collaborative relationship between the MIT Center for Constructive Communication and Cortico. Individuals who were involved in hosting and analyzing the conversations from the NPR and the City of Durham projects were invited to sign up for user interviews as part of this project.

²Prompts and additional system details available in the appendix.

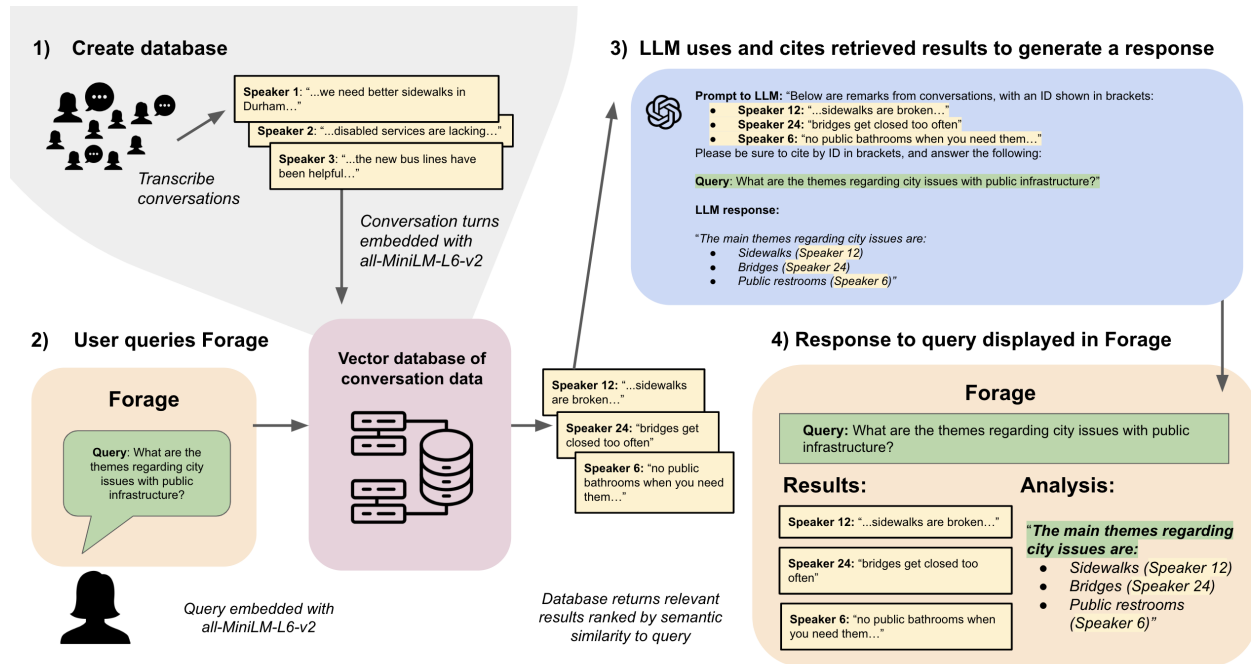


Figure 1: FORAGE 1.0 System diagram

Each user interview provided participants with a guided interaction through FORAGE, which allowed open-ended exploration of their cohort's unique conversation data.

In preparation for user interviews with the NPR cohort, transcripts from the NPR conversation series were uploaded to FORAGE, which were then viewable in FORAGE for the NPR cohort of study participants. Our NPR cohort of 10 unique study participants included journalists and community engagement managers involved in analyzing the community conversations with the goal of creating radio stories that highlighted participant perspectives on the theme of "community." All participants in the NPR cohort had at least some previous experience with AI tools in their work, but they had mixed impressions of and understanding of these tools. They reported mixed regularity of AI usage, and they mentioned a range of tools including ChatGPT, Grammarly, and Atlas.ti. Many also used the analysis platform developed by Cortico to store, view, and annotate the conversation data, which contains some AI-based features, such as annotation suggestions. Participants reported that their current usage of AI tools was often guided by policies set by their workplace regarding appropriate and inappropriate uses of the tools in journalistic settings, including guidelines regarding privacy preservation and maintaining the originality of editorial content.

For the Durham cohort of our study, data from the 70 recorded BeHeard Durham conversations were included in FORAGE for interview participants to explore. Participants in our interview study from the City of Durham included community engagement managers and representatives from individual departments within the city, such as Human Relations or Neighborhood Improvement Services. Individuals from the organization who were involved in organizing, analyzing, or using data from the BeHeard conversation

series in their work were recruited for this study, and 9 unique individuals from this cohort participated. These participants had mixed experience with AI tools, and mixed impressions of AI in general.

4.3 Interview study

To explore needs and user interactions with FORAGE, we conducted semi-structured interviews with a think-aloud protocol, first with 10 unique individuals from NPR, occurring between November and December of 2024. Following our design change from FORAGE 1.0 to FORAGE 2.0, we conducted two follow-up interviews with NPR journalists requesting feedback on the change. Our second user study, focused on FORAGE 2.0, included 9 unique participants from the City of Durham, and occurred in December, 2024 and January, 2025. All participant interviews were hosted for at least 30 minutes.

Our interview protocol had three stages. First, **Situating the user**, we asked participants to share about their current tool use, if they used any AI-based tools in their work, and if they had any concerns about using AI. Second, we guided users through **Interacting with FORAGE**. They were first guided to construct a query on a familiar topic in their data, before constructing a query on an unfamiliar topic or dataset. Finally, a **Reflection** asked the participant to share how they might see this tool being useful, whether and how any interactions raised concerns about reliability or biases of the tool, and any feedback they had on the user interface. The full interview protocol is available in the appendix. We analyzed the interviews iteratively and inductively using a method of inductive thematic analysis [7] inspired by grounded theory [13] to answer our main research questions.

5 Findings

5.1 Users seek synthesis, not just retrieval

In participant interviews, participants spent the vast majority of their time engaging with the "analysis" side of the screen, rather than reviewing the "search" results showing relevant speaker turns on the left of the screen. Participants usually looked at the "analysis" side first, before following links to citations in the search results from the analysis.

Each participant produced at least two queries through FORAGE during their interview study, and we intentionally provided minimal query scaffolding in the first cohort of participant interviews using FORAGE 1.0 to learn more about the queries users made on their data with minimal guidance. We provided several canned example queries in the interface, including *"Summarize positive mentions of sidewalks"* and *"Generate themes about gentrification"*. Participants had access to the tool indefinitely once the interview session completed, and some participants continued using the tool after the interview session concluded as they continued work on their projects.

5.1.1 Taxonomy of RAG queries. In order to understand main use cases, we sought to describe the kinds of queries made by users on the FORAGE platform. Taxonomies of user query types were critical for understanding search and retrieval systems on the early web. Classic web search paradigms describe several query types, including "navigational" queries (to reach a website), "informational" queries (to find or acquire information from a website), and "transactional" queries (to accomplish a transaction, such as downloading a file or buying a ticket) [8, 15]. There are some potential similarities between the kinds of queries users could make on the web and in FORAGE, such as "informational" queries that strictly retrieve information from the corpus. However, informational queries "find information assumed to be available on the web in a static form" [8] whereas RAG-based systems like FORAGE offer the possibility not just of finding information, but also synthesizing it in a way that contributes to the production of new knowledge and insight. RAG-based systems appear to be able to synthesize information through summarization, topic generation, and some forms of thematic analysis. As we could not find any taxonomy or descriptive insight into common types of RAG-based queries for data analysis given these new use cases, we devised a descriptive taxonomy of query types informed by participant interviews and user logs.

The two lead authors devised the taxonomy by manually reviewing participant interviews and user logs, and each proposed a draft taxonomy. The authors met to discuss and converge on taxonomy definitions. One author applied this descriptive taxonomy to each recorded query in order to acquire descriptive labels of the user log history, which totaled 165 queries after discarding duplicate entries made consecutively by the same user. Figure 2 shows the distribution of query types we observed across all participants in the study. **Theme generation** queries, like *"Generate themes related to gentrification"* were the most common type of query, accounting for 36% of the total queries made in our study period. **Retrieval** queries, like *"Find me comments about a generational divide"* were next most common at 27% of the total. **Synthesis** queries included queries like *"Summarize gen z's politics,"* a task requiring synthesis

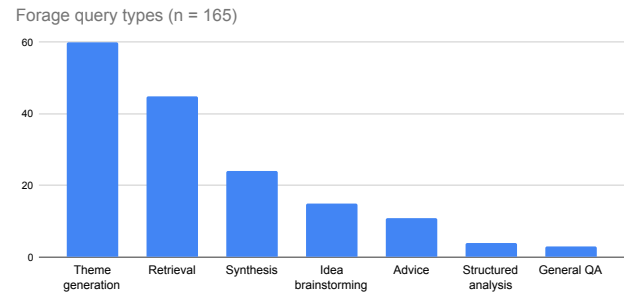


Figure 2: Histogram of annotated query types in FORAGE from 165 user queries.

of information with respect to a request, but not necessarily the generation of interpretive themes. These accounted for 15% of the total. The dominance of **Synthesis** and **Thematic generation**, two related categories, suggests users tended to use the tool's affordances more often for diverse forms of synthesis than for retrieval alone. This insight aligns with emerging work showing researchers' widespread use of LLMs in brainstorming and analysis, tasks which go beyond sorting and retrieving data.

A smaller but noteworthy category includes **Advice** queries (7%) like *"What can be improved in Durham's infrastructure?"*. We note these with interest, because they may indicate a desire to treat FORAGE as an expert system that produces an opinionated analysis, rather than neutrally surfacing existing data or objective patterns. Journalists in our cohort, in particular, commonly used FORAGE for **Idea brainstorming** queries (9%), like *"Generate story pitches about immigrant journeys"*. These may indicate a desire to use FORAGE to generate original content for their professional context, not just surface, sort, summarize data, or derive themes. Story pitches generated with these queries often made a case for a unique angle or insight the story pitch could take, driving the LLM's role beyond synthesis. Finally, we observed a few requests for **Structured analysis** (2%) that would require data about the community in a format the conversation data is unlikely to provide, such as those requiring location data, and some **General QA** (question-answering) queries (2%) making factual questions that need data external to the corpus, such as crime statistics sought by the query *"Which streets have lots of crime?"* We discarded just 2 queries from this taxonomic annotation where the query type was unclear. These were *"perfect"* and *"helpful"*, which may reflect the users' expectation to chat back with our system after receiving results from a query.

The tendency within our participant group to use the tool to produce thematic analysis, interpretation, and original content is striking, and motivates needed follow-up work in both methodology and tool design. This taxonomy can guide discussion and future research about user behavior when querying these systems.

5.2 FORAGE can support user needs for sensemaking and retrieval

Across both participant cohorts, interviews overall revealed excitement that this tool could help participants sort through long hours

of unstructured conversation data ($n = 13$). For journalists, the obvious use case was in finding "tape" from conversations quickly in order to support the creation of compelling radio stories ($n = 7$). Many participants specifically appreciated that quotes raised by FORAGE could be verified with direct links back to the source material, mitigating some concerns journalists had initially raised regarding the risks of generative AI ($n = 6$). Journalists reported FORAGE helped them find stories they had forgotten or did not know about in their own data ($n = 5$). It also helped surface insights from conversations with which they were unfamiliar because they did not host it ($n = 3$). They also found value in the way FORAGE surfaced themes across conversations, exposing users to cross-cutting topics and themes that could inspire story pitches for radio that surfaces recurrent themes across communities ($n = 4$).

City workers found related but subtly different value in the tool, with many sharing it could help find examples of the "human side" of problems to support grantmaking and budgeting proposals ($n = 3$). Others saw potential for the tool to aid departments in understanding problems through conversations hosted outside their own department ($n = 3$).

FORAGE surfaced new insights that users had not already realized in the data, going beyond simply retrieving content or confirming information users already knew. For example, when **P5B** queried FORAGE on the Durham corpus, she noted that FORAGE surfaced "student housing" as a subtheme of the greater subject area of housing concerns, an issue she "wasn't even considering" prior to using FORAGE for sensemaking. She reviewed the cited stories about student housing and learned student housing was a real concern for some conversation participants. FORAGE yielded new insight about subthemes that connected stories across communities in the NPR cohort as well. **P2A** noticed FORAGE surfaced "impact of COVID on third spaces" as a subtheme generated with his query about "third spaces". When FORAGE surfaced this subtheme with citations to comments from community conversations in locations other than **P2A**'s, he mentioned interest in using this subtheme as the basis of follow-up interviews within his own community for a developing news story.

5.3 Open-endedness is a feature and a bug

5.3.1 Open-ended interfaces present challenges. Echoing past work on LLMs in qualitative research [26], participants in our study showed enthusiasm for interacting with data in an open-ended way, but this open-endedness did present several challenges. First, the open-ended nature of the interface resulted in participants approaching FORAGE with some query types that it was not designed to handle. For example, some users appeared to have tried to chat back with the system, or give additional system instructions through the query text box. For example, when interacting with FORAGE 1.0, a user queried "What topics came up in all conversations? Don't consider things said by [name of facilitator]". The interface uses similarity-based retrieval in the first step to retrieve relevant results before a generation step, and this query did not give a particular subject area to retrieve, so it was unable to successfully complete this task. Because FORAGE 1.0 used the entire typed query to create an embedding that would be used to retrieve relevant results, the

additional instructions ("Don't consider things said by [name of facilitator]") unrelated to subject matter created noisier search results when included in the embedding and retrieval step. Furthermore, because speaker name was not included as part of the conversation data, the system was unable to follow these instructions, and the participant was unsatisfied with performance on these results.

We also saw examples of users trying to add additional context to their query, such as this one: "I work for The City Planning Department, which guides growth, zoning, and development to create an inclusive, sustainable, and equitable community. The department manages land use, development proposals, zoning ordinances, and long-range planning efforts to ensure Durham's growth aligns with community and environmental goals. What are people saying about parks and their importance in the community?" In creating queries with additional context or instructions that were not handled, users could also be dissatisfied with generated answers if additional instructions were not honored. One user from the cohort of NPR participants who queried "Do people feel misunderstood?" then queried "Are there patterns in how people feel their communities are misunderstood?" before finally querying "Personal anecdotes or stories about feeling misunderstood". Anecdotally, this may suggest the first two sets of results may not have provided the user with the content they sought, and that they refined their query into formats suggested by the tool in the canned example by the third query.

Simultaneous with reviewing this user behavior, we realized that including the user's **Objective** instructions to the model, such as "Generate themes" in the embedding text for retrieval likely created noise in the retrieved responses, which could be avoided. Furthermore, the general open-endedness of the interface led to confusion about what the system was well-suited to do, prompting us to consider adding additional scaffolding.

These two issues motivated a change in the query interface from FORAGE 1.0 to FORAGE 2.0. In FORAGE 1.0, we included a single text box where any text query could be typed. In FORAGE 2.0, we prompted users to break apart their query into a box capturing the **Objective**, such as "generate themes about" or "find stories related to," and a box capturing the **Subject**, such as "problems with sidewalks" or "voter disenfranchisement." Taken together, an **Objective** and **Subject** build a query of the following form: **Subject:** "find stories related to" + **Object:** "voter disenfranchisement". Only the **Subject** in FORAGE 2.0 is used to create the embedding for retrieval, decreasing noise in search results. The string concatenation of **Subject** + **Object** still instructs the LLM to create synthesis atop the retrieved results. We discuss the results of this change in a later section.

5.3.2 The ambiguity of natural language creates challenges. Some users encountered challenges communicating with the tool due to the inherent ambiguity of natural language. We observe instances where participants deliberated over word choice to refer to a **Subject** area for their query. For example, participant **1A** wanted to find stories about immigrants' journeys to the US, and queried for "themes related to immigrant journeys". She was initially surprised that the results included a theme "mentioning of dreams, of creating groups of communities of immigrants in these new places to support each other" because the results were "not necessarily [about] the journey." However, upon discussion, **P1A** realized the word "journey" was ambiguous, and that maybe "being vague" had

given her results that were "more...metaphorical than the literal," which had surprised her given her expectations of AI technology. The participant refined her query to specify "physical journey," producing results she understood better, but she still continued to edit her query substantially to find the results she sought and was not fully satisfied with performance.

Participants edited their query word choice to improve results (**P2A**, **P1A**, **P8A**), but sometimes participants were not sure if the tool would "understand" the term they wanted to use and they shifted their querying behavior accordingly. Participant **P2A** initially used the term "community spaces" in a query, which surfaced results including "third spaces"—the term he had initially wanted to use. He then edited his initial query to reference to "third spaces" instead of "community spaces," and he said the new results were "almost exactly what [he] was hoping," a surprising outcome, because he was "expecting the program to not necessarily understand" the term "third space."

Participants navigated linguistic ambiguity with respect to the **Objective** they requested as well. Journalists in the NPR cohort (**P8A**, **P9A**) first queried for "stories about [x]" or "themes about [x]", but they found these results did not connect stories to themes for a story pitch of the format they needed. However, they were excited by the results produced by querying FORAGE for "Story pitches about [x]". Several participants used their existing understanding of AI to make sense of FORAGE behaviors and set their expectations accordingly, with **P1A** saying "I know it's only as good as the prompt I give it," and **P9B** saying he knows AI is "garbage in, garbage out."

5.4 Light scaffolds could support appropriate use

Open-ended interaction patterns are flexible but can lead to confusion about expected behavior and system capabilities, providing a challenge for developers aiming to create a predictable model of system behavior [4] and support appropriate, intentional use. Light scaffolds on query construction could help users find what they want, and avoid system failure due to query construction. We tried one small change to our interface from FORAGE 1.0 to 2.0, prompting users to break their queries apart into an **Objective** and **Subject**, minimizing additional instructions and only using the **Subject** for retrieval. We conducted two follow-up interviews with participants **P3A** and **P2A**, who reviewed the change from FORAGE 1.0 to 2.0 and shared positive feedback. They agreed the change made understanding the bounds of possible interactions easier to understand, steering better use of the tool. Reviewing user logs, we also find users nearly strictly adhered to well-formed queries following the change to FORAGE 2.0. The change avoided queries that treated the system as an expert system or oracle, and instead steered users towards queries that surfaced patterns to facilitate their own sensemaking. This change likely had trade-offs for flexibility of interaction, and our implemented change is just one of many changes that could support appropriate use of the tool. Future work can investigate other ways to create light scaffolds for appropriate use.

Beyond creating light scaffolds on user queries in systems like these, onboarding design can better set user expectations about system behavior, but without this first insight into how users wanted

to use the tool, providing recommendations was difficult. Providing recommendations about appropriate use presupposes thorough evaluation of the system's capabilities to determine how the system could perform across objectives, subject domains, and data types, information unknown at the time of this formative study. Building on these insights, the system may need to detect inappropriate or malformed queries or use cases, and respond with clarification or refusal, design choices for the system that could be explored in future work.

5.5 Validation needs may differ by use case

Participants such as **P1A** shared concerns about LLMs' capacity for hallucination, and emphasized their need to ensure accuracy themselves ($n = 4$). Users appreciated that FORAGE results link to the source data, increasing user confidence in findings through reviewing the sources linked in the outputs ($n = 5$). Journalist **P3A** shared "a general hesitation as far as using AI for editorial purposes because of... hallucinations. We, as reporters, are responsible for double and triple checking." Because FORAGE is "linking directly" to the conversation, she said she can "go back and see where it pulled things from, just to make sure that we're comfortable with how it's interpreting things."

However, users may validate outputs from FORAGE differently depending on their relationship to the source material. Validating outputs may be easier when the user knows the source data well, since we observe that users rely on their existing knowledge of the data to validate results ($n = 8$). However, when a user queries FORAGE to make sense of content from conversations they either did not host or have never reviewed themselves outside of FORAGE, their process for validating insights may be different and require additional consideration in our design. Some participants mentioned they would need to spend more time validating LLM-generated outputs by checking content from linked conversations they did not already know well ($n = 3$). The additional time required to validate LLM-generated analysis derived from unfamiliar sources could deter users from properly validating outputs, and supporting appropriate validation should be an area for future design.

6 Future directions and limitations

The distribution of query types we observe in Figure 2 indicates that participants largely seek thematic analysis and synthesis, objectives that do more than retrieve data alone. Our participants were overall impressed by the prototype's performance in this regard, and enthusiastic about using the tool for these purposes. However, we note significant unknowns regarding the use of LLMs to support sensemaking. For these use cases, the LLM produces an interpretation of the data, including producing judgments about which features in the data are prevalent and salient enough to surface, what to call them, and how to justify them.

Given that LLMs are the product of constrained training data and further shaped by alignment practices, LLM-powered sensemaking systems are likely to produce analysis and perspectives that reflect or align with some groups more than others. Depending on the data under analysis, this risks misinterpretation, particularly of minorities or marginalized groups. Our study centered on two US-based user groups, and our findings may not generalize to other

languages, countries, and cultural contexts. For example, one participant tried using the tool to query a subset of Spanish language data using Spanish language queries, and was dissatisfied with the results. This suggests caution is needed in extending this approach to other contexts, and indicates a much-needed area of future work to evaluate performance on diverse sources of data from varied domains.

Furthermore, the mechanism by which these insights are generated by the system is not transparent, an issue worsened by the widespread use of proprietary models [20]. In our case, as for many researchers, using the OpenAI models was convenient due to the product's API and the availability of highly performing models within our budget. For use cases where reproducibility and transparency of mechanism are particularly important, the current system likely does not suffice, and future work should investigate performance of these systems when powered by open source models. Future iterations of the taxonomy should be updated to reflect more diverse user logs, and can help evaluate performance and design for appropriate use.

As discussed in the previous section, additional work is needed to understand users' practices of verifying outputs from the tool, practices which may vary depending on users' relationship to the data under analysis. Including direct citations to sources in the data is reassuring to users who must prioritize faithfulness to primary sources. If users start with minimal knowledge of the data, they may accept LLM interpretations of the data at first glance. Research is showing that LLMs can be homogenizing forces in insight generation [9] and creative tasks [3, 5]. Furthermore, the generated analysis we include in FORAGE produces a single perspective on the analysis objective, which risks reinforcing a particular interpretation of the data given that humans are strongly subject to anchoring bias [28] and uptake biased LLM outputs [3].

Given these risks, FORAGE may not be appropriate for all sense-making contexts or tasks. Organizations and research teams undergoing a sensemaking process should carefully consider which "objectives" are appropriate for an LLM-powered tool to support or complete, and which should be kept free from LLM influence [26]. For example, we note that some queries were made that appear to treat FORAGE as an expert system, rather than simply a tool to surface data that can inform a human's systematic interpretation and conclusion about the data. Concerns about bias and overgeneralization may be enhanced for contexts using the tool to analyze data from minority groups or vulnerable populations. Participants in this study had a range of familiarity with AI tools, and the concerns we raise may be enhanced for users with particular backgrounds or levels of understanding of the risks introduced by AI.

7 Impacts

Participants showed strong interest in using FORAGE for more than retrieval and summarization, frequently using it to inform sense-making through the generation of thematic insight or for brainstorming ideas within their professional context. The participating cohort of NPR journalists used the FORAGE prototype to analyze conversations about community, often in teams of multiple journalists who collaborated across cities in the US. NPR published several national news stories in part facilitated by sensemaking done on

their data through FORAGE. These include published stories about immigrants to the United States [23] and Americans discussing their sense of community [1].

8 Conclusion

Journalists and city workers found FORAGE, a RAG-based tool, helpful for quickly exploring, searching, and interpreting their unstructured text data. They showed strong interest in the LLM-generated analysis compared to pure search, describing the tool as a "game-changer" for going through long hours of conversations and identifying themes across conversations. Purely open-ended interactions popularized by ChatGPT can lead to confusion about what LLM-based systems can and cannot do, but designing light scaffolds can steer users towards appropriate use. Our work identifying user needs and behavior patterns can inform needed follow-up research on LLM performance in these high-stakes community-centered contexts, as well as the design of systems that support appropriate use.

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A Interview information and ethics

At the start of each interview, we obtained verbal consent to record and participants could stop the interview or refuse to answer any question. All interview transcriptions were created through Zoom’s built-in transcription service, or through a locally hosted Whisper model and edited manually by the research team to ensure fidelity to the speaker’s words. We used the Dedoose platform and video review to analyze the interview sessions.

We informed participants that each FORAGE query and set of corresponding results entered into the FORAGE prototype was saved in our database. Participants were told not to share any personal information with the prototype, and that we would be analyzing their interactions with the tool for research purposes. We also informed them an LLM powered the prototype. We did not use model configurations at any stage that train on input data from public conversations or user queries.

B Interview Protocol

Section 1: Situating the user

- Can you give me a high-level overview of your work?
 - How did you become involved in this project? How many conversations have you run or hosted?
 - How did you become involved in this project? How many conversations have you run or hosted?
- What have you learned from the conversations so far?
- What tools do you currently use in your work? (Examples: excel, document searching, MaxQDA, etc)
- Have you used any AI tools in your work so far?

Section 2: Interacting with FORAGE

Then, we transitioned to the hands-on portion of the user interview. Our interview script was: "We will now transition to giving you access to the tool directly. I'll ask you to screen share so that it's you driving the interaction, but I can help you if necessary. Think-Aloud encouragement: As you interact with FORAGE, we want to hear as many of your thoughts, feelings, and reactions as possible. I may interrupt you to remind you to share what you're seeing and anything you're noticing." Then, we shared the link to the prototype and their login information. When they loaded the prototype and shared their screen, we introduced the tool, drawing attention to several features: "You can choose the dataset you want to look at with this drop-down menu. There are some pre-loaded queries you can see here. When you are ready to run a query, you type it in here and press this button, then it generates the results."

Then, we scaffolded the interaction for users by calling back to the information shared in the first section of the protocol, where participants shared whether they ran conversations or not, and what insights they had learned from the conversations to that point: "First, we would like you to think about the themes you noticed in your conversations. Let's select the conversation set you were involved with." Then, we encouraged users to create a query in FORAGE related to the dataset with which they were already familiar, say, about a topic or issue that had come up in the conversations they ran.

As the participant sent their first query, we said "The results will pop up with quotes on the left and any analysis requested on the right." As participants saw their first set of results, we responded

to their reaction, choosing from some of the following questions if prompts were needed to encourage the participants to share: "Take a minute to look over the results.

- What do you notice?
- What are you drawn to? (Quotes or analysis?)
- Did this work how you expected?
- Do the results match your memory of the topic in the conversation?
- Does this make you see something in your data differently? How?
- What does this make you wonder next?

The final question was used to steer users towards their next query for the tool.

The second phase of the hands-on interaction with the tool encouraged NPR participants to explore the whole collection of data in contrast to just the data from their own member station, and the second phase with the City of Durham participants encouraged participants to search a topic they had not explored in the corpus to that point.

Depending on the amount of time and interest participants had, they could continue querying the tool and sharing thoughts, but at least 1 query of each type was completed before transitioning to the third and final section of the study.

Section 3: Reflection

For NPR and the City of Durham, we asked a series of similar questions, including

- What is one way you think this might be useful?
- Anything you want to explore more?
- Are there any confusing things about the interface that don't make sense?
- Is there anything you want to make sure LLMs do not do for you as you analyze this conversation data?

- What is one thing that you have doubts about? Are there ways you will avoid using this tool?
- Do you trust the analysis that was generated? Why or why not?

Section 4: Wrap up Finally, we encouraged participants to share feedback with us through a live, linked form in the prototype.

C System details

Following guidance outlined in Pang et al., we include additional details to describe our LLM-powered system's configuration. The embedding model used to create the conversation turn vector database was all-MiniLM-L6-v2 from HuggingFace. Vectors were retrieved using FAISS [11]. FORAGE users have the option to use GPT-3 or GPT-4 to create LLM-augmented analysis above the search results, and users can configure the number of search results to retrieve.

The text used to prompt begins: *"Below are remarks from conversations, with an ID shown in brackets:"*. This is followed by a bulleted list of speaker turns from the retrieved search results, each preceded by a unique numeric identifier in brackets. Following the list of speaker terms, the following text is added: *"Please be sure to cite by ID in brackets, and answer the following:"* Finally, instructions specific to the user's query are appended. In the case of FORAGE 1.0, this is the simply the full text of the query entered by the user. In the case of FORAGE 2.0, it is a resolved template that depends on the objective selected and the subject entered. The available objectives and the templates used for them are shown in Table 1.

D Additional figures

A system diagram of FORAGE 2.0, depicted in Figure 3 is available for comparison to FORAGE 1.0, which is visible in the main body of the work in Figure 1.

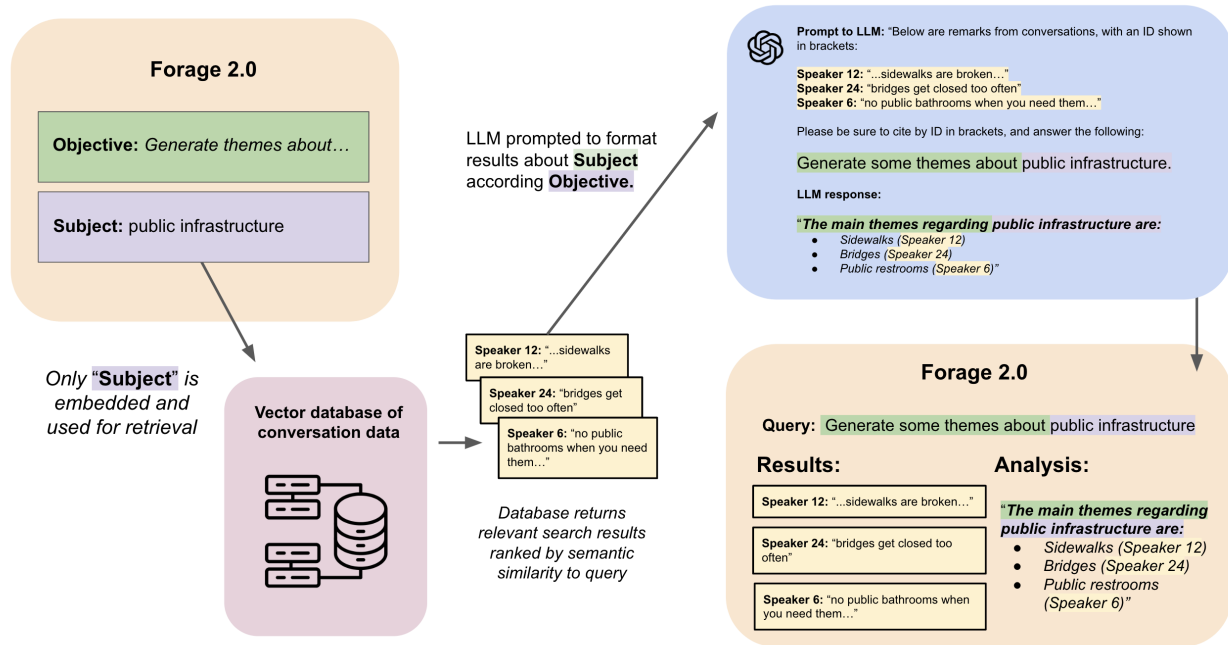


Figure 3: FORAGE 2.0 System diagram. Query is decomposed into two text boxes, one for the Objective and one for the Subject. Only the Subject is used for retrieval, then both the Objective and Subject are used in the LLM generation step.

Table 1: Prompt templates used in FORAGE 2.0

Objective	Prompt template
Generate themes about...	What are themes about [SUBJECT] in these remarks?
Find positive mentions of...	What are positive things people say about [SUBJECT]?
Find negative mentions of...	What are negative things people say about [SUBJECT]?
Summarize opinions about...	Please summarize the opinions people are expressing about [SUBJECT].
Summarize personal experiences shared about...	Please summarize the personal stories people are sharing about [SUBJECT].
Generate story pitches about...	Please generate some story pitches related to the topic of "[SUBJECT]" based on what people are expressing in these remarks.
Find the most memorable quotes about...	Find the most memorable quotes about the topic of "[SUBJECT]" from these remarks.
Other objective	[SUBJECT]

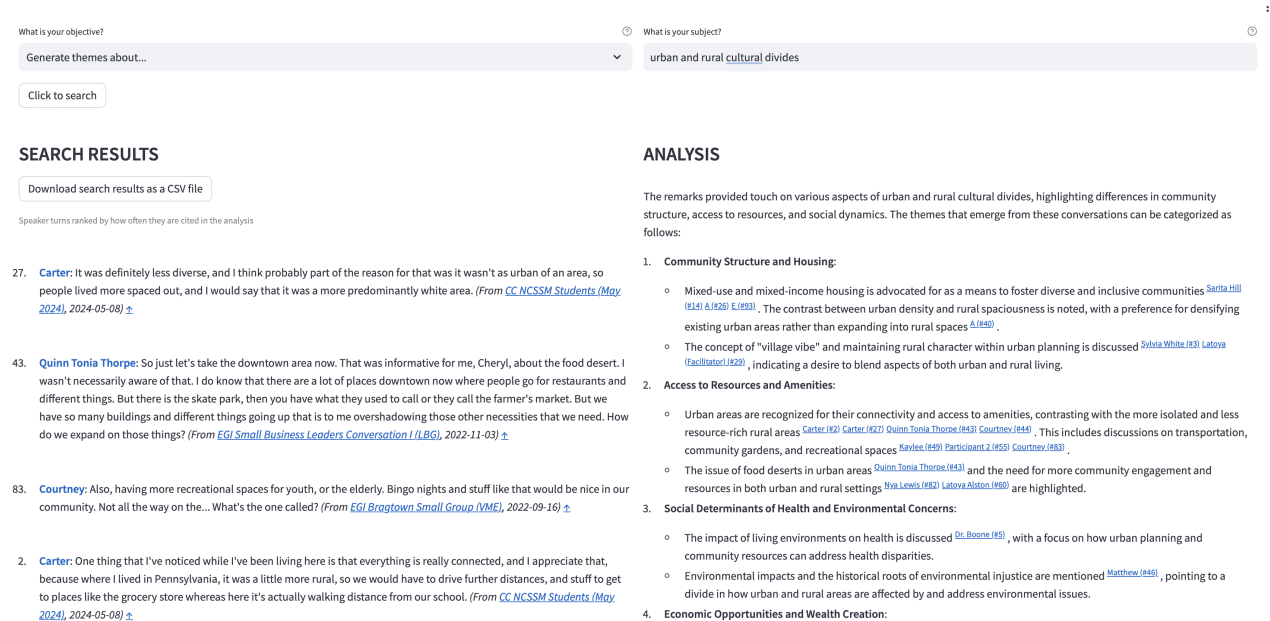


Figure 4: Example query on FORAGE, showing search results on left, and LLM-generated analysis on right, in response to the query "Generate themes about..." (objective) + "urban and rural cultural divides" (subject)